

Supplementary Figure 1B, SpatialPCA Results

The following analysis uses Visium V2 data from Sample P2 in the paper High-definition spatial transcriptomic profiling of immune cell populations in colorectal cancer (<https://doi.org/10.1038/s41588-025-02193-3>).

```
library(rhdf5)

library(ggplot2)
library(fs)
library(glue)

library(SpatialPCA)
```

Load the processed Data

Load the processed Visium data used in Figure 1.

```
counts <- read_h5ad_matrix(pth_proc_adata)
```

```
## The count matrix has dimensions 18035 x 4269
```

```
var <- read_h5ad_meta(pth_proc_adata, "/var")
obs <- read_h5ad_meta(pth_proc_adata, "/obs")

nobs <- nrow(obs)
if (which(dim(counts) == nobs) == 1L) {
  counts <- t(counts)
}
```

Use same genes as in Figure 1

```
# genes3k <- read_csv("../data/visium_genes3k.csv")
genes3k <- readr::read_csv("../../visium_genes3k.csv", col_names=F, col_select=1)
genes3k <- genes3k$X1
```

Define spatialPCA object

```
spa <- new("SpatialPCA")
spa@counts <- counts[genes3k,]
spa@normalized_expr <- as.matrix(spa@counts)
spa@location <- as.matrix(obs[,c("X", "Y")])
```

Run spatial PCA

First, compute kernel

```
spa <- SpatialPCA_buildKernel(
  spa,
  sparseKernel = TRUE
)
```

```
## ## Selected kernel type is: gaussian
## Gene 8 : sample is too sparse to find TD
## Gene 19 : sample is too sparse to find TD
## Gene 24 : sample is too sparse to find TD
## Gene 85 : sample is too sparse to find TD
## Gene 98 : sample is too sparse to find TD
## Gene 111 : sample is too sparse to find TD
## Gene 137 : sample is too sparse to find TD
## Gene 142 : sample is too sparse to find TD
## Gene 195 : sample is too sparse to find TD
## Gene 246 : sample is too sparse to find TD
## Gene 247 : sample is too sparse to find TD
## Gene 248 : sample is too sparse to find TD
## Gene 249 : sample is too sparse to find TD
## Gene 251 : sample is too sparse to find TD
## Gene 253 : sample is too sparse to find TD
## Gene 268 : sample is too sparse to find TD
## Gene 280 : sample is too sparse to find TD
## Gene 289 : sample is too sparse to find TD
## Gene 305 : sample is too sparse to find TD
## Gene 309 : sample is too sparse to find TD
## Gene 322 : sample is too sparse to find TD
## Gene 334 : sample is too sparse to find TD
## Gene 398 : sample is too sparse to find TD
## Gene 403 : sample is too sparse to find TD
## Gene 411 : sample is too sparse to find TD
## Gene 427 : sample is too sparse to find TD
## Gene 435 : sample is too sparse to find TD
## Gene 447 : sample is too sparse to find TD
## Gene 448 : sample is too sparse to find TD
```

```
## Gene 460 : sample is too sparse to find TD
## Gene 467 : sample is too sparse to find TD
## Gene 469 : sample is too sparse to find TD
## Gene 517 : sample is too sparse to find TD
## Gene 524 : sample is too sparse to find TD
## Gene 527 : sample is too sparse to find TD
## Gene 567 : sample is too sparse to find TD
## Gene 571 : sample is too sparse to find TD
## Gene 607 : sample is too sparse to find TD
## Gene 668 : sample is too sparse to find TD
## Gene 699 : sample is too sparse to find TD
## Gene 714 : sample is too sparse to find TD
## Gene 721 : sample is too sparse to find TD
## Gene 730 : sample is too sparse to find TD
## Gene 742 : sample is too sparse to find TD
## Gene 743 : sample is too sparse to find TD
## Gene 745 : sample is too sparse to find TD
## Gene 747 : sample is too sparse to find TD
## Gene 765 : sample is too sparse to find TD
## Gene 767 : sample is too sparse to find TD
## Gene 774 : sample is too sparse to find TD
## Gene 779 : sample is too sparse to find TD
## Gene 781 : sample is too sparse to find TD
## Gene 791 : sample is too sparse to find TD
## Gene 801 : sample is too sparse to find TD
## Gene 816 : sample is too sparse to find TD
## Gene 828 : sample is too sparse to find TD
## Gene 841 : sample is too sparse to find TD
## Gene 843 : sample is too sparse to find TD
## Gene 860 : sample is too sparse to find TD
## Gene 863 : sample is too sparse to find TD
## Gene 887 : sample is too sparse to find TD
## Gene 892 : sample is too sparse to find TD
## Gene 920 : sample is too sparse to find TD
## Gene 922 : sample is too sparse to find TD
## Gene 924 : sample is too sparse to find TD
## Gene 926 : sample is too sparse to find TD
## Gene 941 : sample is too sparse to find TD
## Gene 974 : sample is too sparse to find TD
## Gene 975 : sample is too sparse to find TD
## Gene 1039 : sample is too sparse to find TD
## Gene 1072 : sample is too sparse to find TD
## Gene 1132 : sample is too sparse to find TD
## Gene 1133 : sample is too sparse to find TD
## Gene 1163 : sample is too sparse to find TD
## Gene 1173 : sample is too sparse to find TD
## Gene 1199 : sample is too sparse to find TD
```

```
## Gene 1224 : sample is too sparse to find TD
## Gene 1229 : sample is too sparse to find TD
## Gene 1247 : sample is too sparse to find TD
## Gene 1260 : sample is too sparse to find TD
## Gene 1267 : sample is too sparse to find TD
## Gene 1285 : sample is too sparse to find TD
## Gene 1293 : sample is too sparse to find TD
## Gene 1296 : sample is too sparse to find TD
## Gene 1307 : sample is too sparse to find TD
## Gene 1323 : sample is too sparse to find TD
## Gene 1339 : sample is too sparse to find TD
## Gene 1368 : sample is too sparse to find TD
## Gene 1375 : sample is too sparse to find TD
## Gene 1427 : sample is too sparse to find TD
## Gene 1428 : sample is too sparse to find TD
## Gene 1490 : sample is too sparse to find TD
## Gene 1507 : sample is too sparse to find TD
## Gene 1509 : sample is too sparse to find TD
## Gene 1530 : sample is too sparse to find TD
## Gene 1556 : sample is too sparse to find TD
## Gene 1583 : sample is too sparse to find TD
## Gene 1593 : sample is too sparse to find TD
## Gene 1606 : sample is too sparse to find TD
## Gene 1621 : sample is too sparse to find TD
## Gene 1640 : sample is too sparse to find TD
## Gene 1647 : sample is too sparse to find TD
## Gene 1648 : sample is too sparse to find TD
## Gene 1650 : sample is too sparse to find TD
## Gene 1679 : sample is too sparse to find TD
## Gene 1697 : sample is too sparse to find TD
## Gene 1703 : sample is too sparse to find TD
## Gene 1732 : sample is too sparse to find TD
## Gene 1757 : sample is too sparse to find TD
## Gene 1834 : sample is too sparse to find TD
## Gene 1838 : sample is too sparse to find TD
## Gene 1871 : sample is too sparse to find TD
## Gene 1876 : sample is too sparse to find TD
## Gene 1895 : sample is too sparse to find TD
## Gene 1960 : sample is too sparse to find TD
## Gene 1973 : sample is too sparse to find TD
## Gene 1997 : sample is too sparse to find TD
## Gene 2007 : sample is too sparse to find TD
## Gene 2020 : sample is too sparse to find TD
## Gene 2035 : sample is too sparse to find TD
## Gene 2056 : sample is too sparse to find TD
## Gene 2081 : sample is too sparse to find TD
## Gene 2088 : sample is too sparse to find TD
```

```
## Gene 2098 : sample is too sparse to find TD
## Gene 2107 : sample is too sparse to find TD
## Gene 2110 : sample is too sparse to find TD
## Gene 2113 : sample is too sparse to find TD
## Gene 2121 : sample is too sparse to find TD
## Gene 2154 : sample is too sparse to find TD
## Gene 2156 : sample is too sparse to find TD
## Gene 2162 : sample is too sparse to find TD
## Gene 2175 : sample is too sparse to find TD
## Gene 2193 : sample is too sparse to find TD
## Gene 2197 : sample is too sparse to find TD
## Gene 2244 : sample is too sparse to find TD
## Gene 2249 : sample is too sparse to find TD
## Gene 2254 : sample is too sparse to find TD
## Gene 2255 : sample is too sparse to find TD
## Gene 2300 : sample is too sparse to find TD
## Gene 2311 : sample is too sparse to find TD
## Gene 2345 : sample is too sparse to find TD
## Gene 2378 : sample is too sparse to find TD
## Gene 2390 : sample is too sparse to find TD
## Gene 2397 : sample is too sparse to find TD
## Gene 2420 : sample is too sparse to find TD
## Gene 2422 : sample is too sparse to find TD
## Gene 2433 : sample is too sparse to find TD
## Gene 2439 : sample is too sparse to find TD
## Gene 2451 : sample is too sparse to find TD
## Gene 2456 : sample is too sparse to find TD
## Gene 2470 : sample is too sparse to find TD
## Gene 2481 : sample is too sparse to find TD
## Gene 2489 : sample is too sparse to find TD
## Gene 2526 : sample is too sparse to find TD
## Gene 2561 : sample is too sparse to find TD
## Gene 2566 : sample is too sparse to find TD
## Gene 2621 : sample is too sparse to find TD
## Gene 2640 : sample is too sparse to find TD
## Gene 2649 : sample is too sparse to find TD
## Gene 2658 : sample is too sparse to find TD
## Gene 2660 : sample is too sparse to find TD
## Gene 2706 : sample is too sparse to find TD
## Gene 2719 : sample is too sparse to find TD
## Gene 2724 : sample is too sparse to find TD
## Gene 2738 : sample is too sparse to find TD
## Gene 2751 : sample is too sparse to find TD
## Gene 2754 : sample is too sparse to find TD
## Gene 2765 : sample is too sparse to find TD
## Gene 2769 : sample is too sparse to find TD
## Gene 2826 : sample is too sparse to find TD
```

```
## Gene 2897 : sample is too sparse to find TD
## Gene 2928 : sample is too sparse to find TD
## Gene 2929 : sample is too sparse to find TD
## Gene 2933 : sample is too sparse to find TD
## Gene 2937 : sample is too sparse to find TD
## Gene 2944 : sample is too sparse to find TD
## Gene 2948 : sample is too sparse to find TD
## Gene 2960 : sample is too sparse to find TD
## Gene 2962 : sample is too sparse to find TD
## Gene 2974 : sample is too sparse to find TD
## Gene 2977 : sample is too sparse to find TD
## Gene 2983 : sample is too sparse to find TD
## Gene 2989 : sample is too sparse to find TD
## Gene 2995 : sample is too sparse to find TD
## ## The bandwidth is: 0.0486129202344738
## ## Calculating sparse kernel matrix
## ## Finished calculating kernel matrix.
```

Now, run method

```
spa <- SpatialPCA_EstimateLoading(
  spa, fast=TRUE, SpatialPCnum=20)
```

```
## [1] "Low rank approximation!"
## [1] "Small sample, using top 195 eigenvectors and eigenvalues in the Kernel matrix!"
```

```
spa <- SpatialPCA_SpatialPCs(spa, fast=TRUE)
```

```
## [1] "Small sample, using top 195 eigenvectors and eigenvalues in the Kernel matrix!"
```

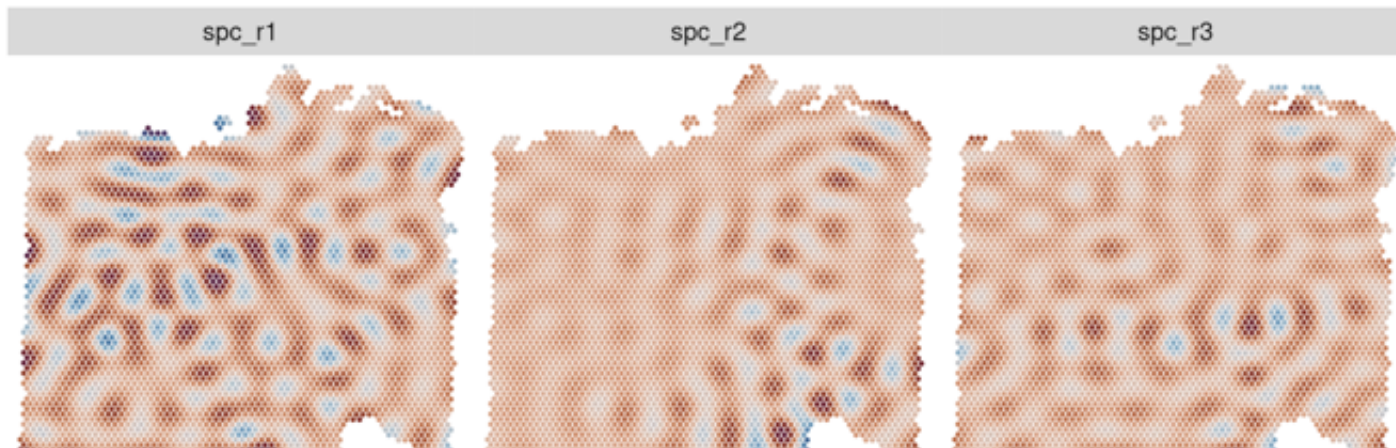
Aggregate results for plotting

```
spcs <- t(spa@SpatialPCs)
colnames(spcs) <- paste0("spc_r", seq(20))
obs <- cbind(obs, spcs)
```

Plot results

```
# Update for each spatial PC
pl_obs <- obs |>
  dplyr::select(X, Y, spc_r1:spc_r3) |>
  tidyr::pivot_longer(
    contains("spc"),
    names_to = "component",
    values_to = "score"
  )

ggplot(pl_obs, aes(X, -1*Y, color=score)) +
  geom_point(size=rel(0.25), alpha=0.90) +
  facet_wrap(~ component) +
  scale_x_continuous(expand=c(0.03,0.03)) +
  scale_y_continuous(expand=c(0.03,0.03)) +
  scico::scale_color_scico(
    breaks = function(x) range(x),
    labels = c("Low", "High"),
    palette = 'vik0', direction=1,
  ) +
  theme_krpca()
```



Session information

```
sessionInfo()
```

```
## R version 4.5.1 (2025-06-13)
## Platform: x86_64-pc-linux-gnu
## Running under: CentOS Stream 8
##
## Matrix products: default
## BLAS: /opt/R/4.5.1/lib64/R/lib/libRblas.so
## LAPACK: /opt/R/4.5.1/lib64/R/lib/libRlapack.so; LAPACK version 3.12.1
##
## locale:
##  [1] LC_CTYPE=en_US.UTF-8      LC_NUMERIC=C
##  [3] LC_TIME=en_US.UTF-8      LC_COLLATE=en_US.UTF-8
##  [5] LC_MONETARY=en_US.UTF-8  LC_MESSAGES=en_US.UTF-8
##  [7] LC_PAPER=en_US.UTF-8     LC_NAME=C
##  [9] LC_ADDRESS=C             LC_TELEPHONE=C
```

```

## [11] LC_MEASUREMENT=en_US.UTF-8 LC_IDENTIFICATION=C
##
## time zone: America/Los_Angeles
## tzcode source: system (glibc)
##
## attached base packages:
## [1] stats      graphics  grDevices  utils      datasets  methods   base
##
## other attached packages:
## [1] RSpectra_0.16-2  SpatialPCA_1.3.0 glue_1.8.0      fs_1.6.6
## [5] ggplot2_4.0.1    rhdf5_2.54.1
##
## loaded via a namespace (and not attached):
## [1] RColorBrewer_1.1-3      rstudioapi_0.17.1      jsonlite_2.0.0
## [4] umap_0.2.10.0          magrittr_2.0.4         spatstat.utils_3.2-1
## [7] farver_2.1.2           rmarkdown_2.30        vctrs_0.7.1
## [10] ROCR_1.0-11            spatstat.explore_3.6-0 askpass_1.2.1
## [13] CompQuadForm_1.4.4     htmltools_0.5.9       SPARK_1.1.1
## [16] Rhdf5lib_1.32.0        sass_0.4.10           sctransform_0.4.3
## [19] parallelly_1.46.1     pracma_2.4.6          KernSmooth_2.23-26
## [22] bslib_0.9.0           htmlwidgets_1.6.4     ica_1.0-3
## [25] plyr_1.8.9            plotly_4.11.0         zoo_1.8-15
## [28] cachem_1.1.0          igraph_2.2.0          mime_0.13
## [31] lifecycle_1.0.5       iterators_1.0.14      pkgconfig_2.0.3
## [34] Matrix_1.7-4          R6_2.6.1              fastmap_1.2.0
## [37] fitdistrplus_1.2-4    future_1.69.0         shiny_1.12.1
## [40] digest_0.6.39         patchwork_1.3.2       Seurat_5.4.0
## [43] tensor_1.5.1          irlba_2.3.5.1         pdist_1.2.1
## [46] labeling_0.4.3        progressr_0.18.0      spatstat.sparse_3.1-0
## [49] httr_1.4.7            polyclip_1.10-7       abind_1.4-8
## [52] compiler_4.5.1        bit64_4.6.0-1         withr_3.0.2
## [55] doParallel_1.0.17     S7_0.2.1              fastDummies_1.7.5
## [58] MASS_7.3-65           openssl_2.3.4         tools_4.5.1
## [61] lmtest_0.9-40         otel_0.2.0            httpuv_1.6.16
## [64] future.apply_1.20.1   goftest_1.2-3         nlme_3.1-168
## [67] rhdf5filters_1.22.0   promises_1.5.0        grid_4.5.1
## [70] Rtsne_0.17           cluster_2.1.8.1       reshape2_1.4.5
## [73] generics_0.1.4       gtable_0.3.6          spatstat.data_3.1-9
## [76] tzdb_0.5.0           tidyr_1.3.2           hms_1.1.4
## [79] data.table_1.18.0     sp_2.2-0              spatstat.geom_3.6-1
## [82] RcppAnnoy_0.0.23     ggrepel_0.9.6         RANN_2.6.2
## [85] foreach_1.5.2        pillar_1.11.1         stringr_1.6.0
## [88] vroom_1.7.0          spam_2.11-3           RcppHNSW_0.6.0
## [91] later_1.4.5          splines_4.5.1         dplyr_1.2.0
## [94] lattice_0.22-7       bit_4.6.0             survival_3.8-3
## [97] deldir_2.0-4         tidyselect_1.2.1      miniUI_0.1.2
## [100] pbapply_1.7-4        knitr_1.51            gridExtra_2.3

```

```
## [103] scattermore_1.2      xfun_0.56      matrixStats_1.5.0
## [106] stringi_1.8.7         lazyeval_0.2.2 yaml_2.3.12
## [109] evaluate_1.0.5        codetools_0.2-20 tibble_3.3.1
## [112] cli_3.6.5            uwot_0.2.4     matlab_1.0.4.1
## [115] xtable_1.8-4         reticulate_1.44.1 jquerylib_0.1.4
## [118] dichromat_2.0-0.1    Rcpp_1.1.1     globals_0.18.0
## [121] spatstat.random_3.4-3 png_0.1-8      spatstat.univar_3.1-6
## [124] parallel_4.5.1       readr_2.1.6    dotCall64_1.2
## [127] listenv_0.10.0       viridisLite_0.4.2 scales_1.4.0
## [130] ggribges_0.5.7       crayon_1.5.3   SeuratObject_5.3.0
## [133] purrr_1.2.1         scico_1.5.0    rlang_1.1.7
## [136] formatR_1.14        cowplot_1.2.0
```